Chemicals



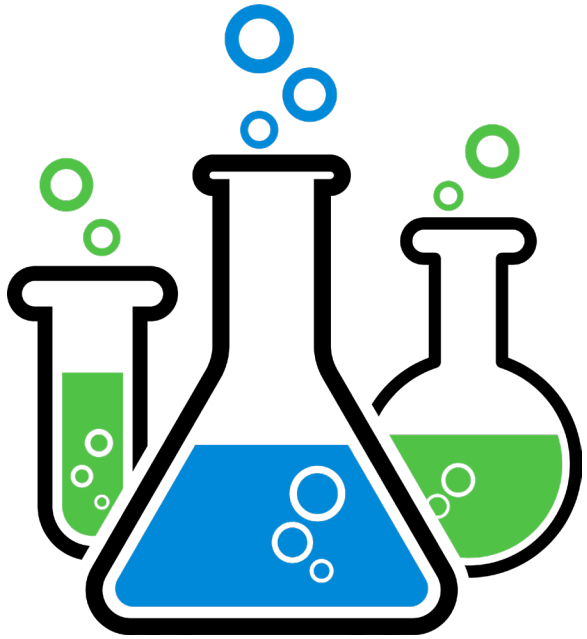
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What is a chemical?

- A chemical is any substance with a defined chemical composition that has specific physical and chemical properties.

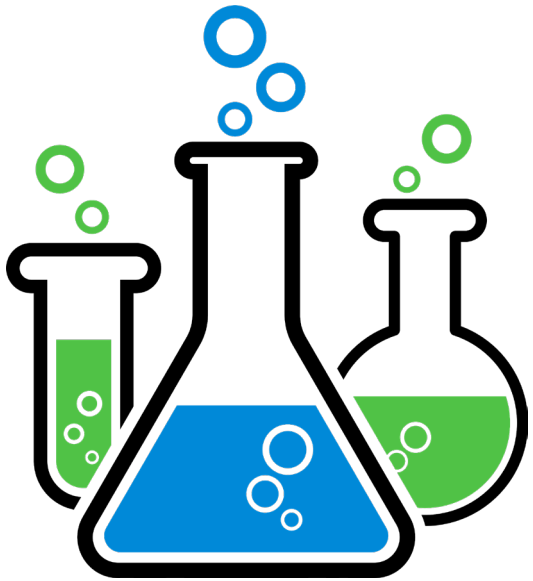
- Chemicals can be:
  - Elements (e.g. copper, hydrogen)
  - Compounds (e.g. water, sodium chloride)
  - Mixtures (e.g. air, alloys, salt solution)

- A chemical's structure determines how it behaves:
  - Electrical conductivity
  - Reactivity with other substances
  - Solubility in liquids



Why chemicals are important

- Engineering systems rely on chemical behaviour to function correctly
- Metals, polymers, ceramics, and composites are all chemicals
- Chemical structure affects:
  - Strength and durability
  - Corrosion resistance
  - Electrical conductivity
- Batteries and cells rely on chemical reactions to produce electricity
  - Electrolytes must be chemically suitable and soluble
  - Poor chemical selection increases internal resistance and heat

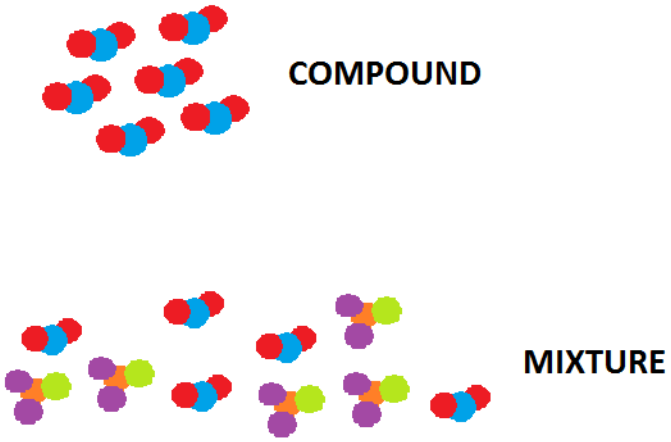


Types of Chemicals: Compound

- A compound is a chemical made from two or more elements **chemically bonded** together in a fixed ratio.

- **Key features**
  - **Fixed composition**
  - New properties different from the original elements
  - Cannot be separated by physical means

- **Engineering examples**
  - Water (H_2O) – coolant, cleaning
  - Sodium chloride (NaCl) – electrolytes
  - Silicon dioxide (SiO_2) – glass, insulation



Types of Chemicals: Compound

- Chemical Formula: Compounds are represented by their chemical formula.
- A chemical formula is a symbolic representation of the proportions of atoms that constitute a particular chemical compound.

Empirical Formula

C_2H_5

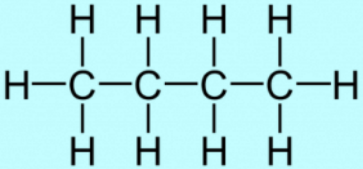
simplest whole number ratio

Molecular Formula

C_4H_{10}

number and type of atoms

Structural Formula



graphic representation of structure

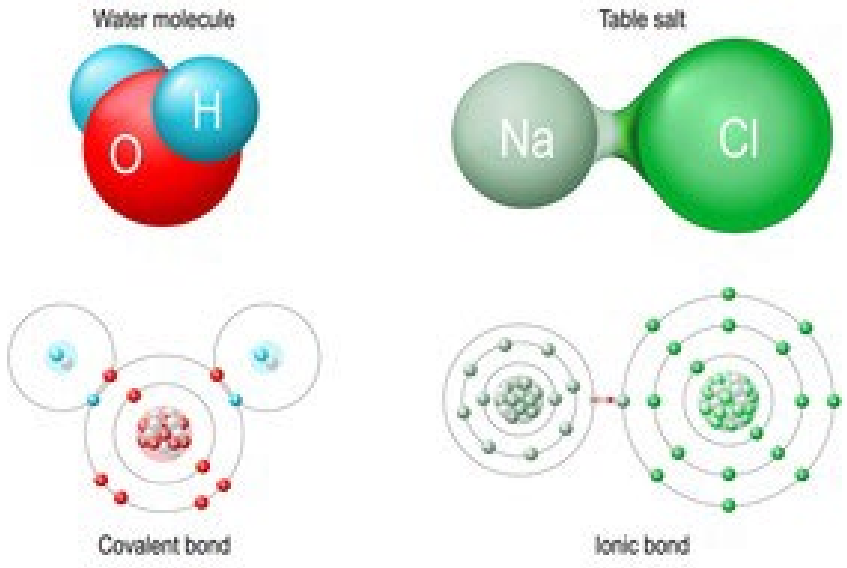
Condensed Formula

$CH_3CH_2CH_2CH_3$

order and formula of functional groups

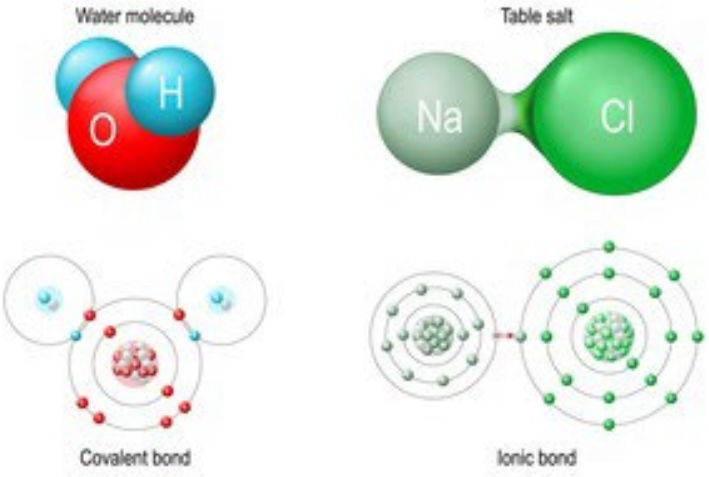
Types of Chemicals: Compound

- Compounds can be classified into two types, **molecular compounds** and **salts**:
  - In molecular compounds, the atom binds each other through **covalent bonds**. (when electrons are shared between atoms)
  - In salts, it is held together with **ionic bonds**. (when electrons are transferred from one atom to another)



Why structure matters in engineering

- Bond type affects conductivity (ionic vs covalent)
- Molecular shape affects solubility
- Functional groups affect reactivity and corrosion
- Structure affects internal resistance in electrolytes

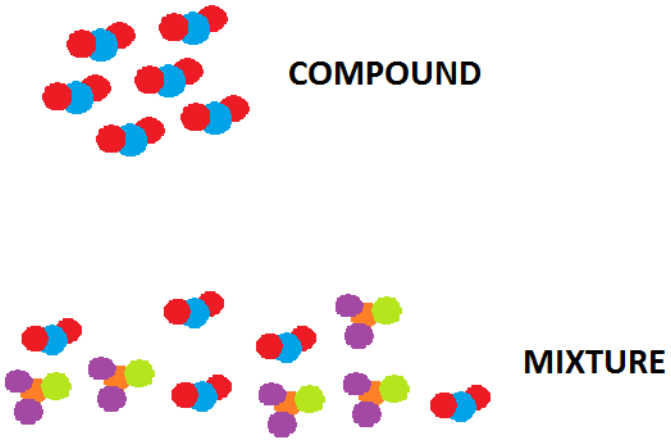


Types of Chemicals: Mixture

- A mixture is a combination of substances not chemically bonded.

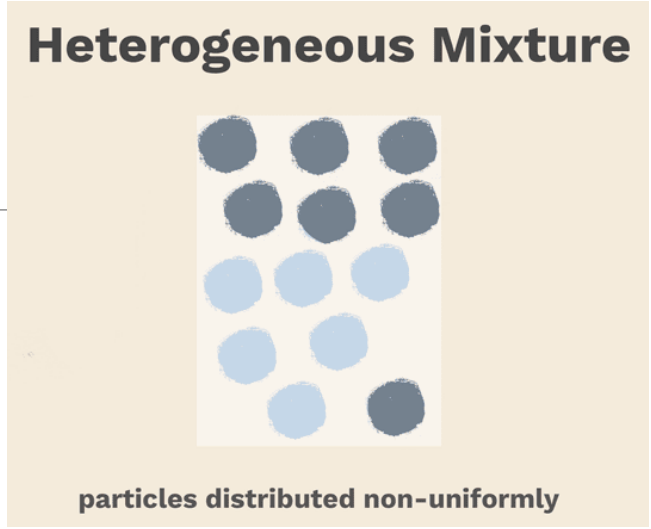
- **Key features**
  - Variable composition
  - Components retain their properties
  - Can be separated by physical methods

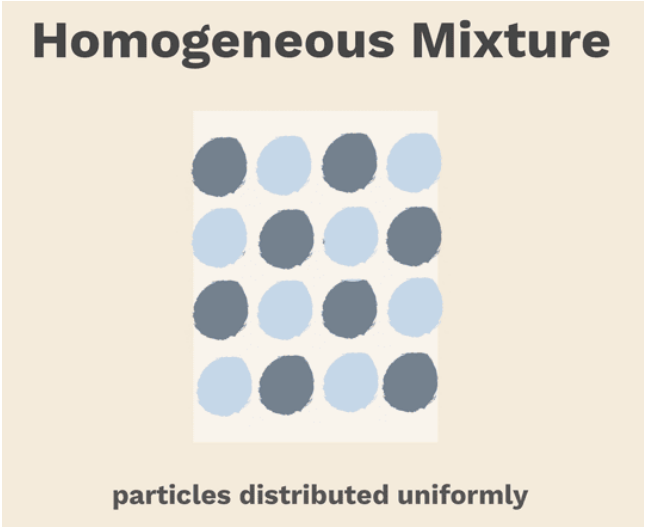
- **Engineering examples**
  - Alloys (e.g. steel, brass)
  - Air
  - Oil with additives
  - Salt dissolved in water



Types of Chemicals: Mixture

- There are two main types of mixtures:
- **Homogeneous** solutions are solutions with **uniform composition** and properties throughout the solution. For example a cup of coffee, perfume, cough syrup, a solution of salt or sugar in water, etc.
- **Heterogeneous** solutions are solutions with **non-uniform composition** and properties throughout the solution. A solution of oil and water, water and chalk powder and solution of water and sand, etc.

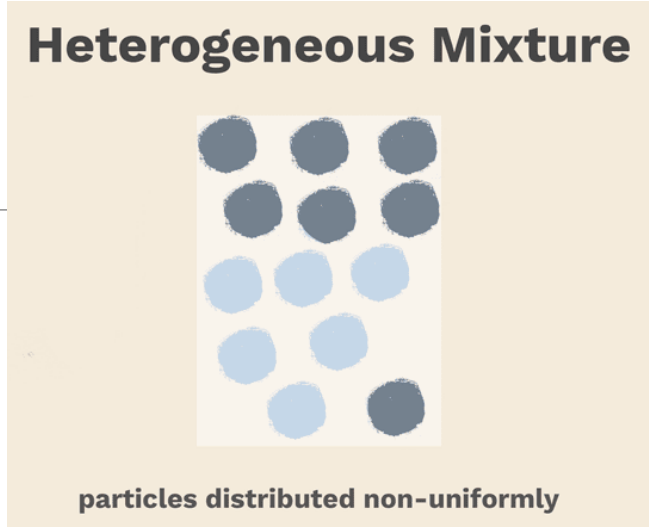


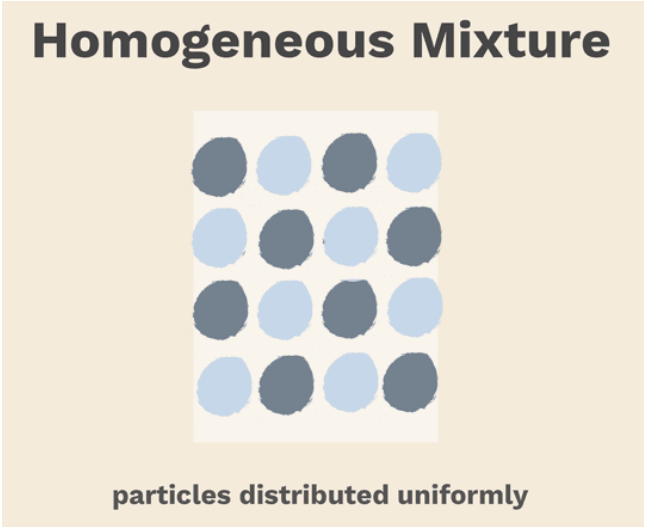


Types of Chemicals: Solution

- A solution is a specific type of **Homogeneous** mixture where:
  - A solute is dissolved in a solvent
  - Particles are at the molecular or ionic scale
  - The solute will not separate out

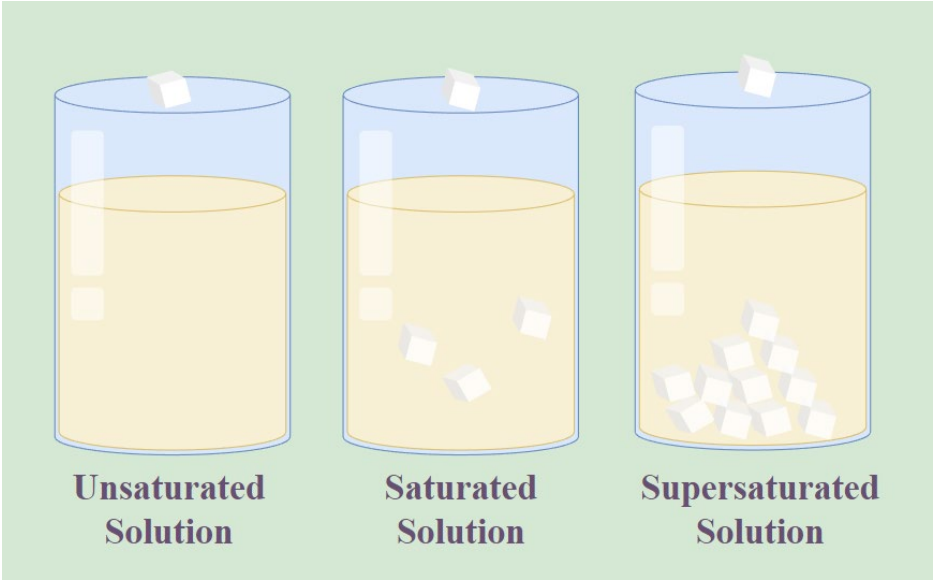
- Examples
  - Salt dissolved in water
  - Sugar in water
  - Electrolytes in batteries





Types of Solutions – Saturation

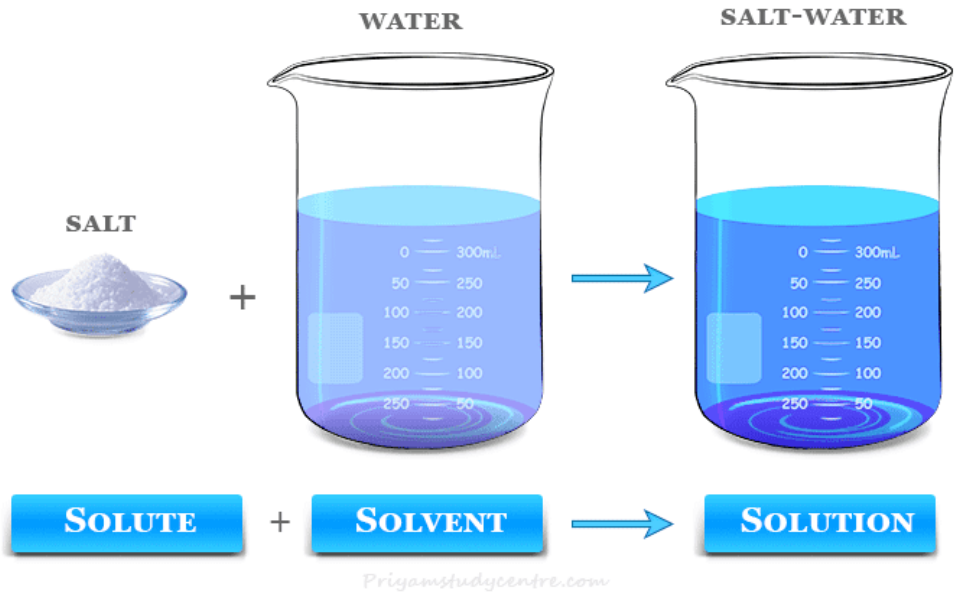
- **A supersaturated solution** comprises a large amount of solute at a temperature wherein it will be reduced, as a result the extra solute will crystallize quickly.
- **An unsaturated solution** is a solution in which a solvent can dissolve more solute at a given temperature.
- **A saturated solution** can be defined as a solution in which a solvent cannot dissolve any more solute at a given temperature.



Types of Solutions – Aqueous

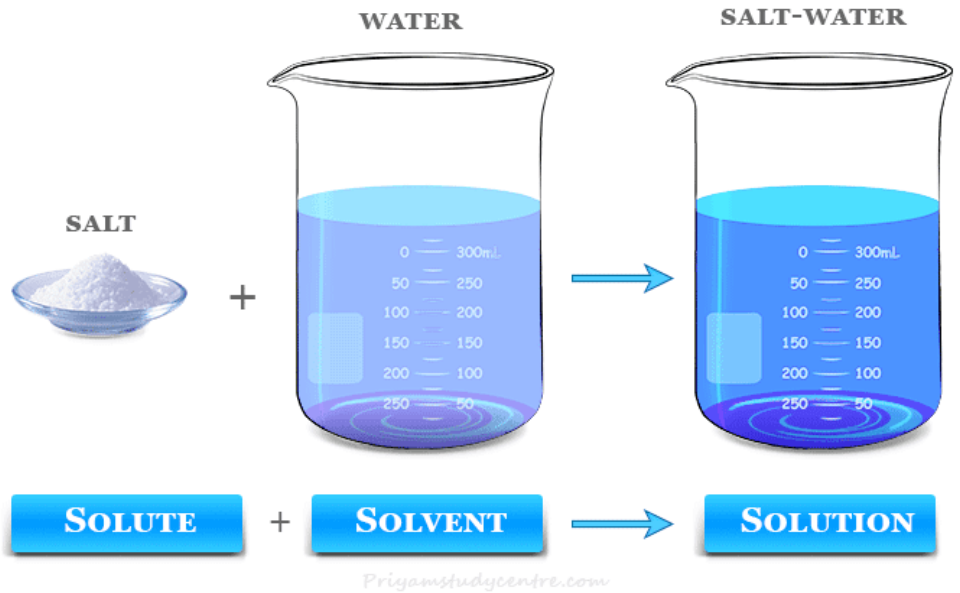
- **Aqueous solution** – When a solute is dissolved in water the solution is called an aqueous solution. Eg, salt in water, sugar in water and copper sulphate in water.

- **Non-aqueous solution** – When a solute is dissolved in a solvent other than water, it is called a non-aqueous solution. Eg, iodine in carbon tetrachloride, sulphur in carbon disulfide, phosphorus in ethyl alcohol.



Types of Solutions – Concentration

- **A dilute solution** contains a small amount of solute in a large amount of solvent.
- **A concentrated solution** contains a large amount of solute dissolved in a small amount of solvent.



Types of Chemicals: Suspension

- A suspension is a specific type of **Heterogeneous** mixture where:
  - Particles with a diameter greater than 1000 nm such that the particles are visible to naked eyes.
  - The particles of a suspension do not pass through a filter paper. So a suspension can be separated by filtration.
  - The suspension is unstable. The particles of a suspension settle down after some time.

- Examples
  - Muddy water
  - Sand particles suspended in water
  - Flour in water

